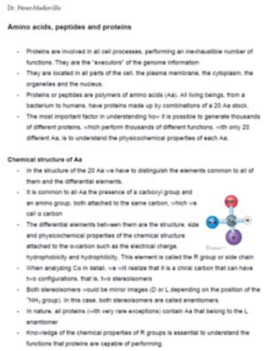
UNIT 2.
Lecture 2.2. Protein structure I



Amino acids, peptides and proteins

- Proteins are involved in all cell processes, performing an inexhaustible number of functions. They are the "executors" of the genome information
- They are located in all parts of the cell, the plasma membrane, the cytoplasm, the organelles and the nucleus.
- Proteins or peptides are polymers of amino acids (Aa). All living beings, from a bacterium to humans, have proteins made up by combinations of a 20 Aa stock.
- The most important factor in understanding how it is possible to generate thousands of different proteins, which perform thousands of different functions, with only 20 different Aa, is to understand the physicochemical properties of each Aa.

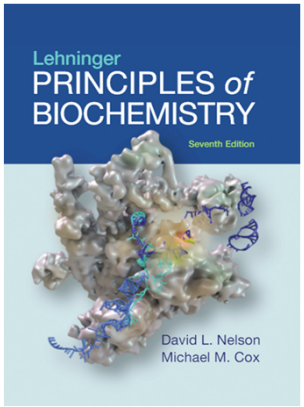
Chemical structure of Aa

- In the structure of the 20 Aa we have to distinguish the elements common to all of them and the differential elements.
- It is common to all Aa the presence of a carboxyl group and an amino group, both attached to the same carbon, which we call α carbon
- The differential elements between them are the structure, size and physicochemical properties of the chemical structure attached to the α-carbon such as the electrical charge, hydrophobicity and hydrophilicity. This element is called the R group or side chain
- When analyzing Cα in detail, we will realize that it is a chiral carbon that can have two configurations, that is, two stereoisomers
- Both stereoisomers would be mirror images (D or L depending on the position of the "NH2 group). In this case, both stereoisomers are called enantiomers.
- In nature, all proteins (with very rare exceptions) contain Aa that belong to the L enantiomer
- Knowledge of the chemical properties of R groups is essential to understand the functions that proteins are capable of performing.

Additional study material

ADI document:

Protein structure. Lecture Notes



Reading

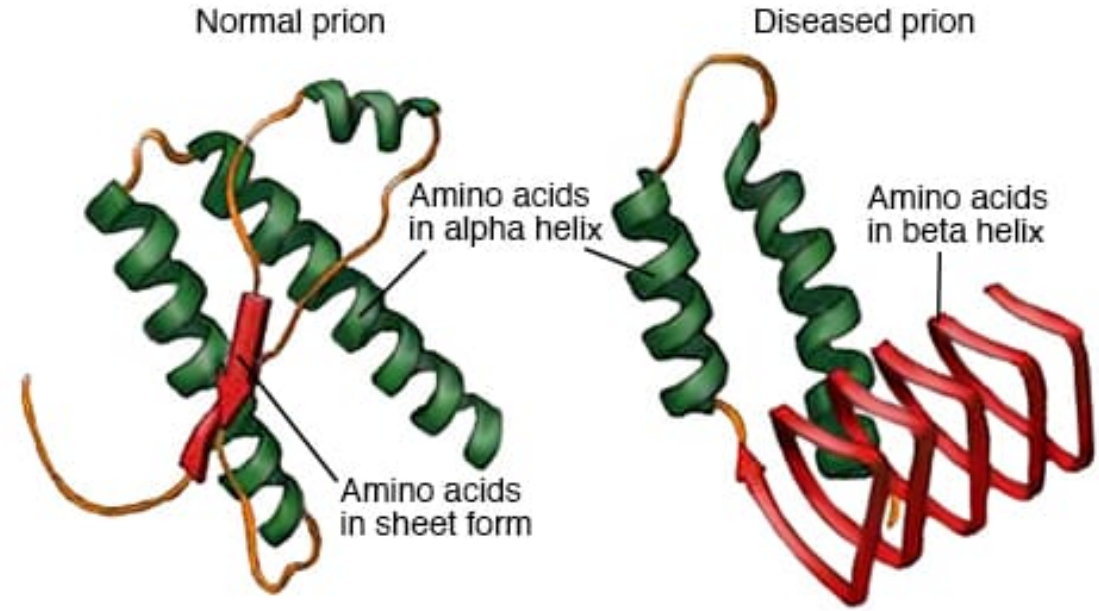
Lehninger's PoB Chapters 3, 4, 5:

Amino Acids and Peptides

Learning goals:

- Structure and naming of amino acids
- Structure and properties of peptides
- Ionization behavior of amino acids and peptides

Protein misfolding = human disease
The case of Prions



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Proteins:
Main Agents of Biological Function

- Catalysis
  - enolase (in the glycolytic pathway)
  - DNA polymerase (in DNA replication)
- Transport
  - hemoglobin (transports O_2 in the blood)
  - lactose permease (transports lactose across the cell membrane)
- Structure
  - collagen (connective tissue)
  - keratin (hair, nails, feathers, horns)
- Motion
  - myosin (muscle tissue)
  - actin (muscle tissue, cell motility)
- Intracellular communication
  - small G proteins (Ras)
  - kinases (MEK)

Amino Acids: Building Blocks of Protein

- Proteins are linear heteropolymers of α-amino acids.

- Amino acids have properties that are well suited to carry out a variety of biological functions:
  - capacity to polymerize
  - useful acid-base properties
  - varied physical properties
  - varied chemical functionality

Amino Acids Share Many Features, Differing Only at the R Substituent

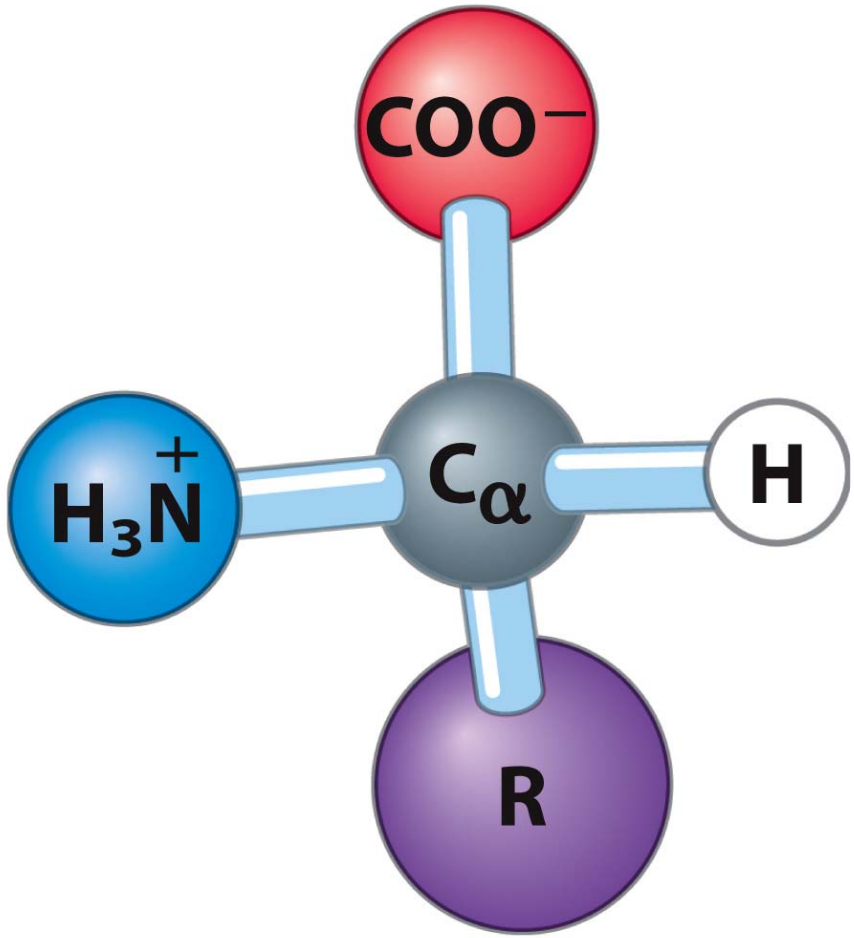


Figure 3-2
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Amino Acids Have Three Common Functional Groups Attached to the α Carbon

- The α carbon always has four substituents and is tetrahedral.
- All (except proline) have:
  - an acidic carboxyl group connected to the α carbon
  - a basic amino group connected to the α carbon
  - an α hydrogen connected to the α carbon
- The fourth substituent (R) is unique in glycine, the simplest amino acid. The fourth substituent is also hydrogen.

All Amino Acids Are Chiral (Except Glycine)

Proteins only contain L amino acids

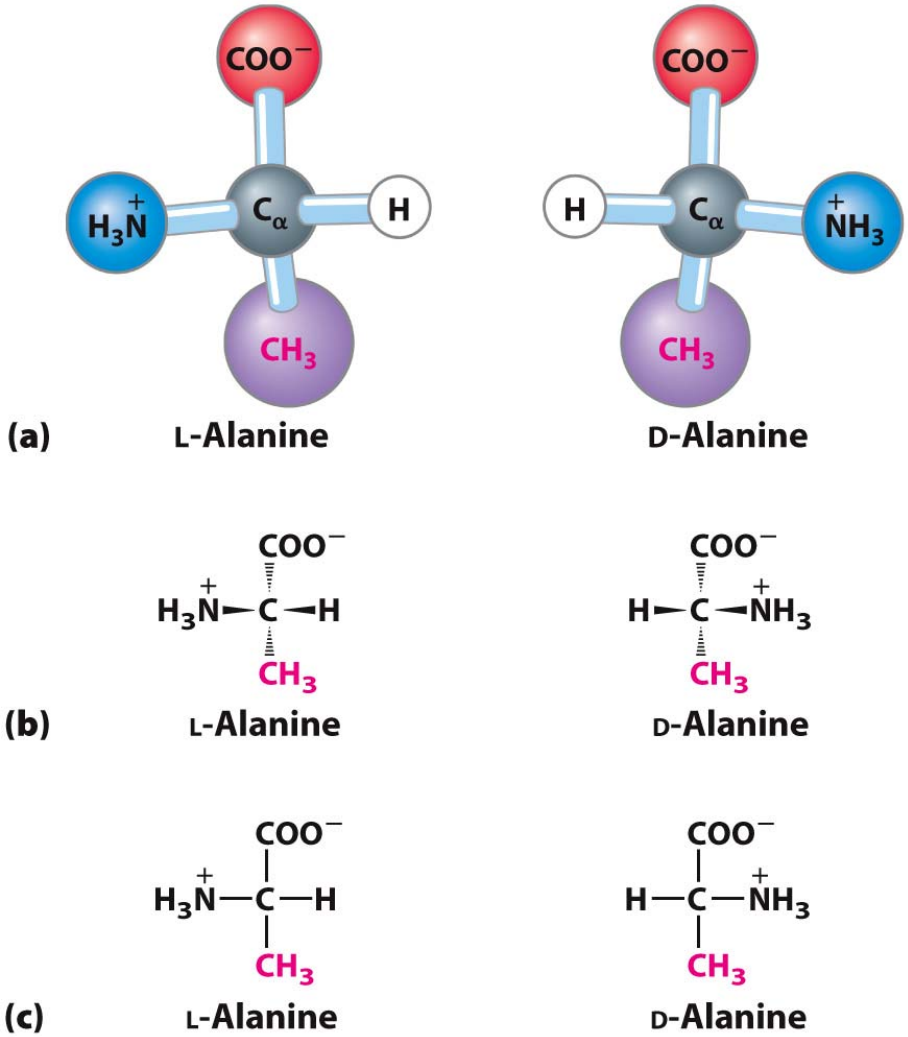


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Amino Acids: Classification

Common amino acids can be placed in five basic groups depending on their R substituents:

- nonpolar, aliphatic (7)
- aromatic (3)
- polar, uncharged (5)
- positively charged (3)
- negatively charged (2)

Nonpolar, aliphatic R groups

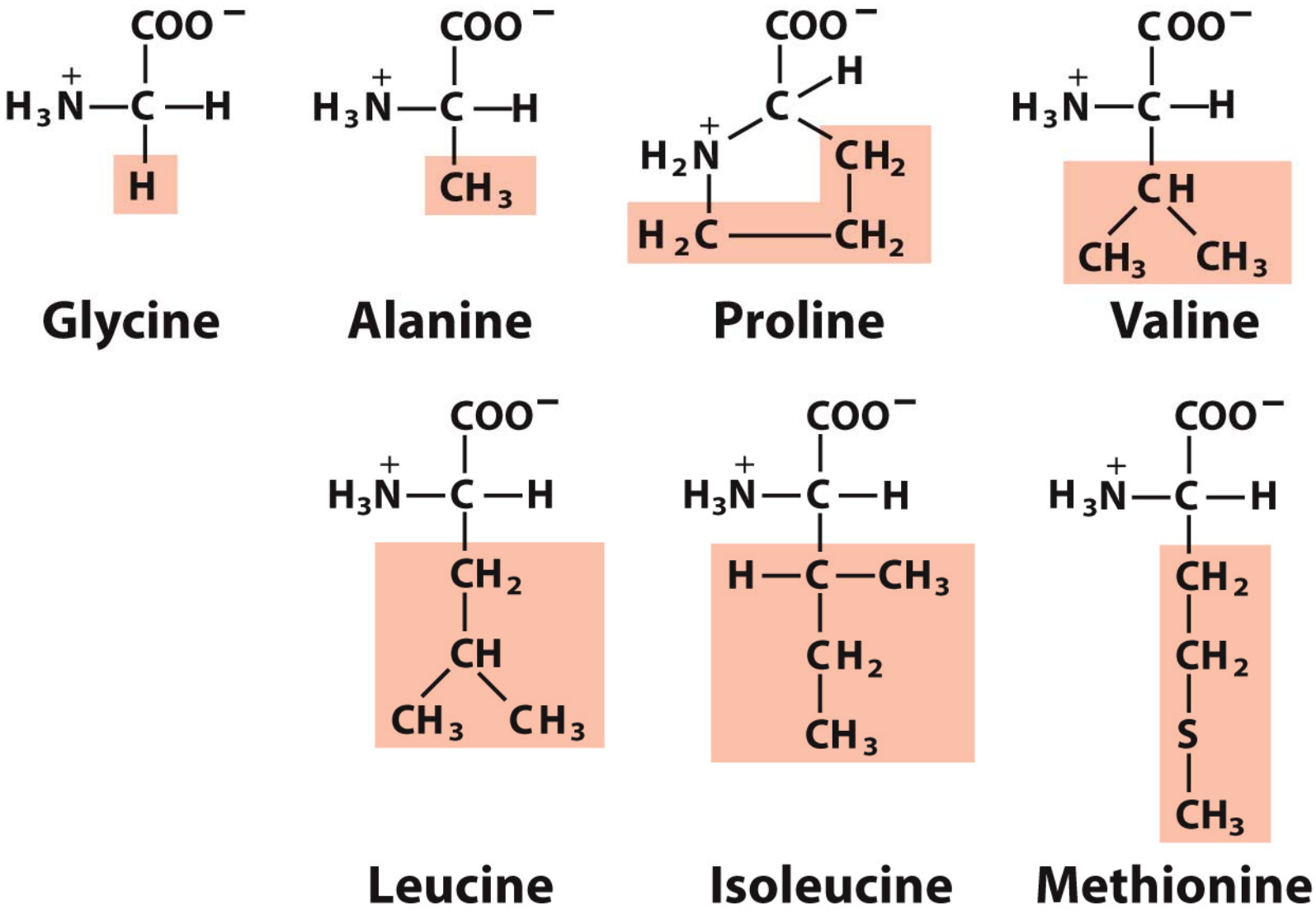
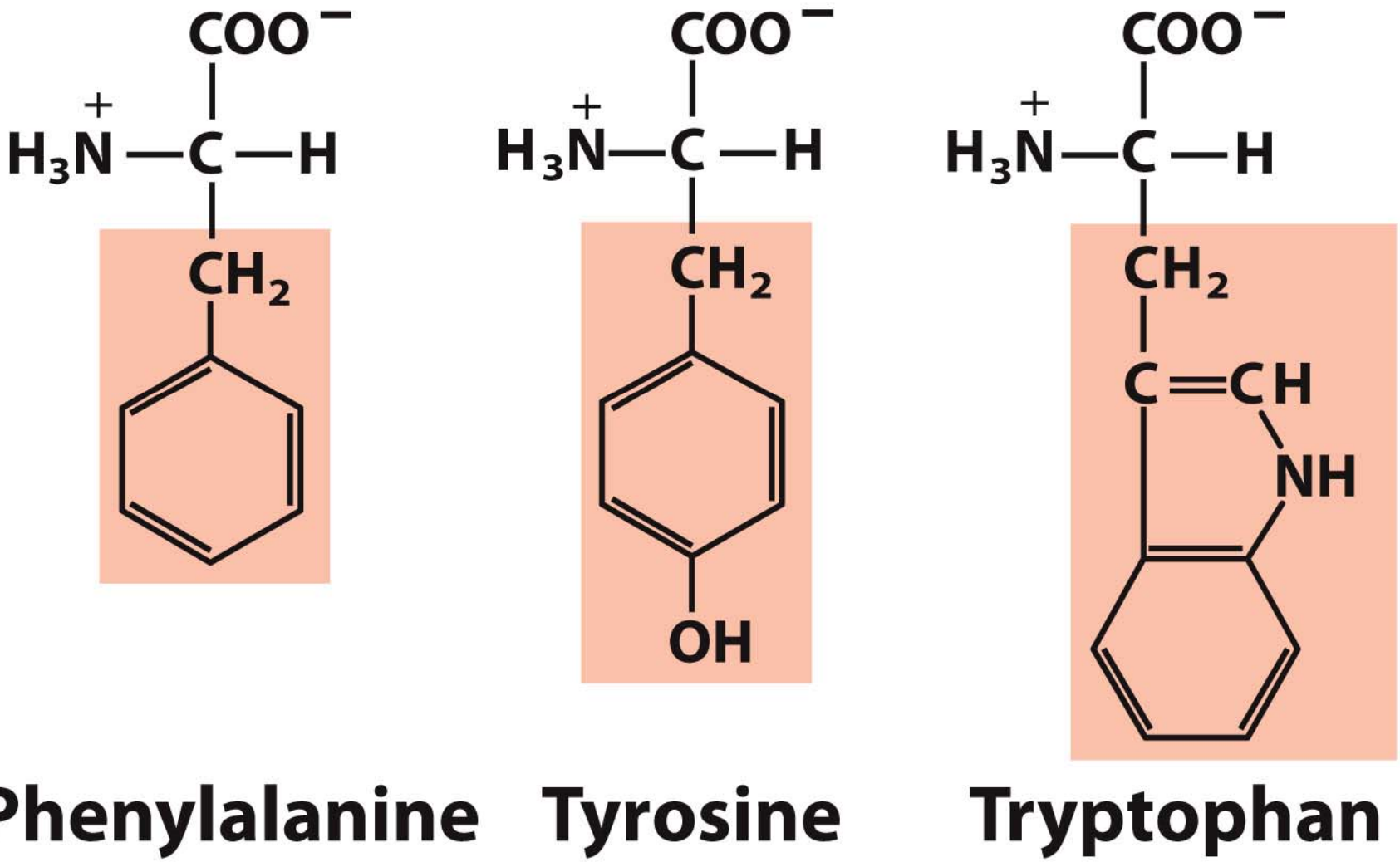


Figure 3-5 part 1
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Aromatic R groups



Phenylalanine **Tyrosine** **Tryptophan**

Figure 3-5 part 2
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Aromatic Amino Acids Absorb Light in a Concentration-Dependent Manner

- The aromatic amino acids absorb light in the UV region.
- Proteins typically have UV absorbance maxima around 275–280 nm.
- Tryptophan and tyrosine are the strongest chromophores.
- For proteins and peptides with known extinction coefficients (or sequences), concentration can be determined by UV-visible spectrophotometry using the Lambert-Beer law: $A = \varepsilon c l$

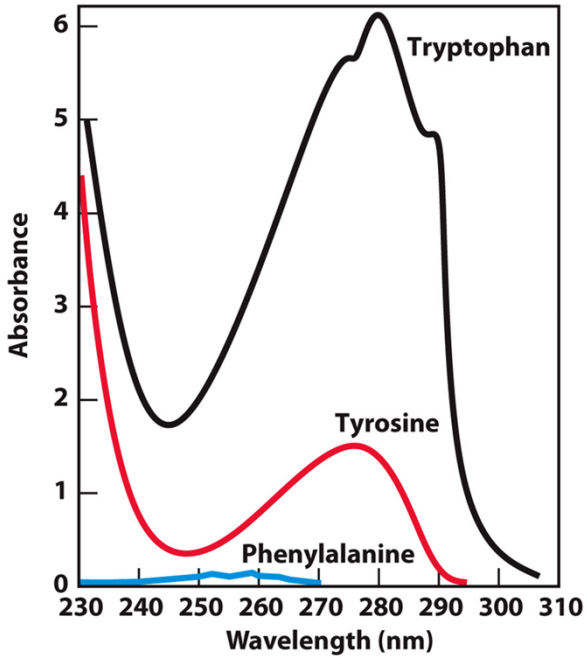


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Polar, uncharged R groups

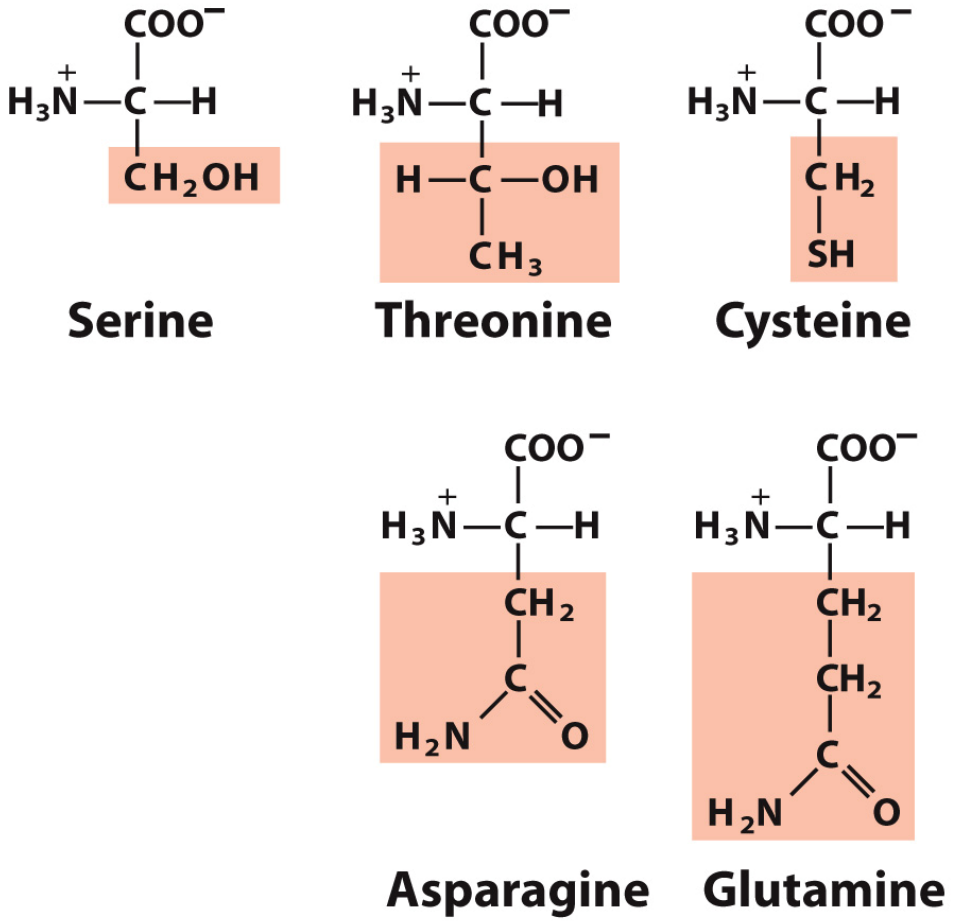


Figure 3-5 part 3
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These amino acids side chains can form hydrogen bonds.
Cysteine can form disulfide bonds.

Positively charged R groups

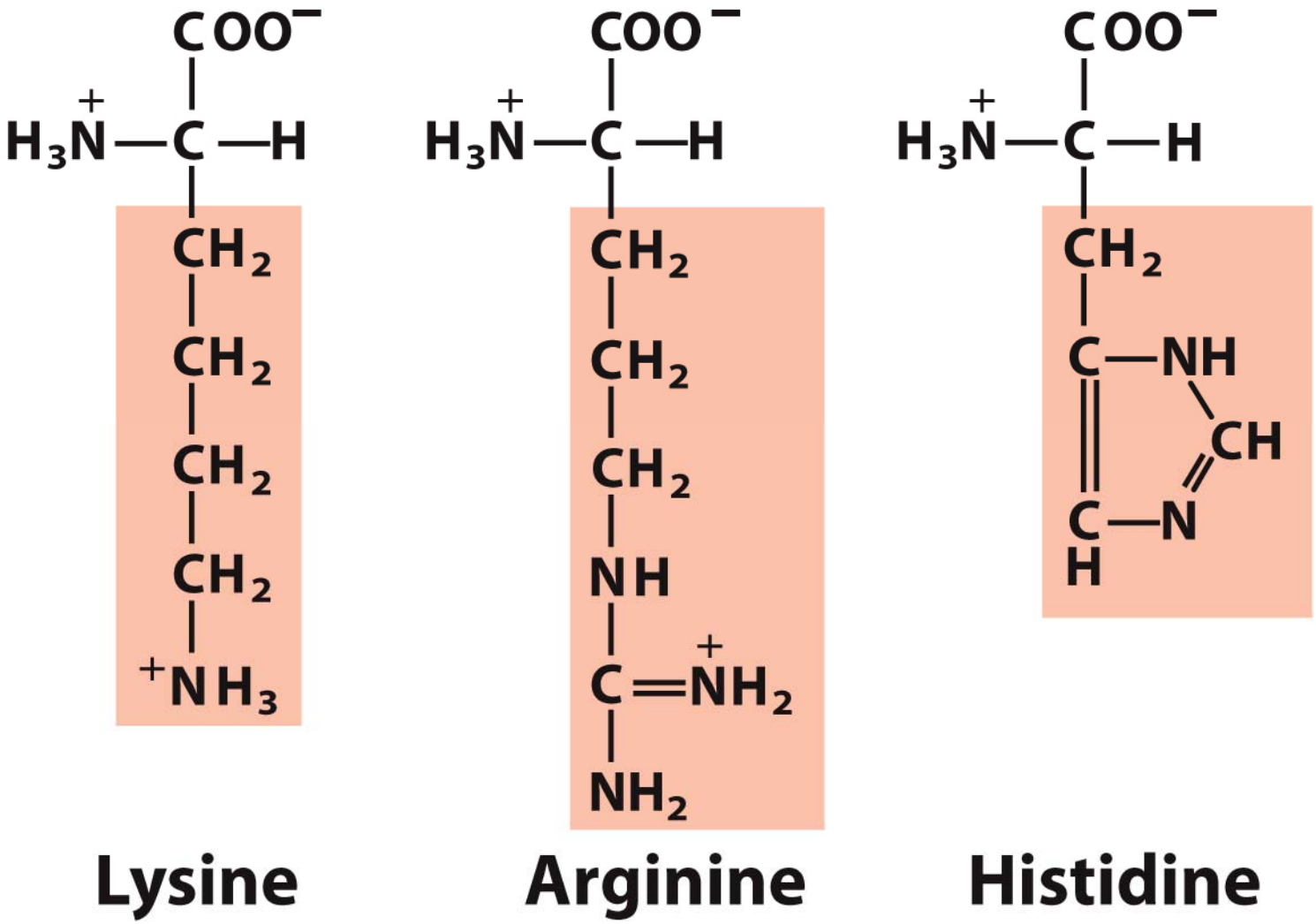
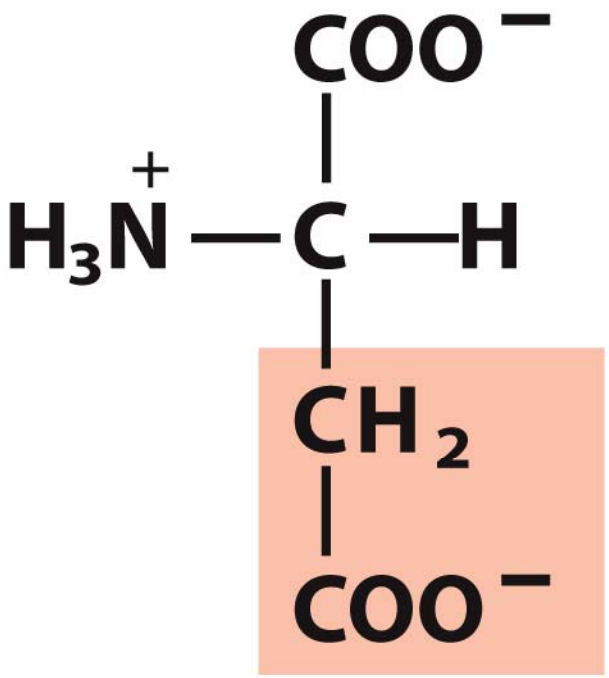
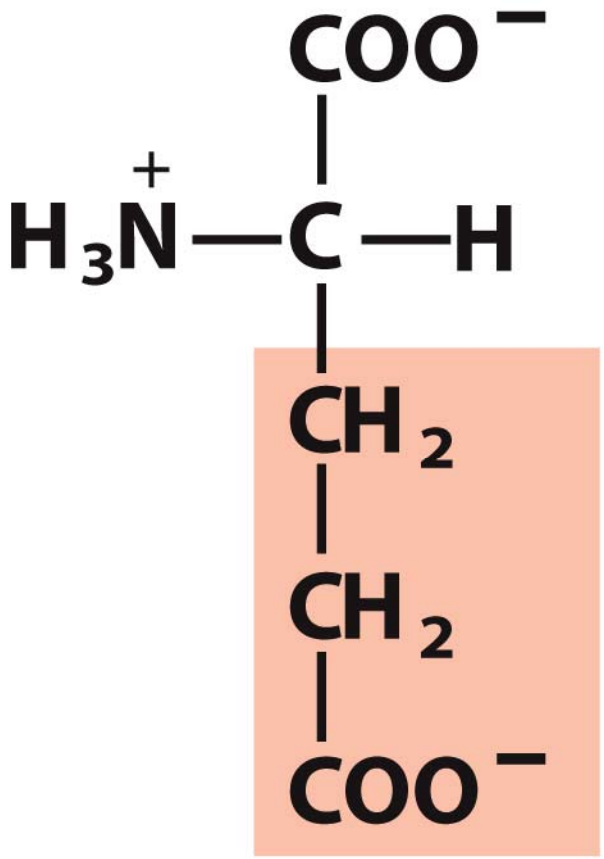


Figure 3-5 part 4
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Negatively charged R groups



Aspartate



Glutamate

Figure 3-5 part 5
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Ionization of Amino Acids

- Amino acids contain at least two ionizable protons, each with its own pK_a.
- The carboxylic acid has an acidic pK_a and will be protonated at an acidic (low) pH:

$$-COOH \leftrightarrow COO^- + H^+$$

- The amino group has a basic pK_a and will be protonated until basic pH (high) is achieved:

$$-NH_4^+ \leftrightarrow NH_3 + H^+$$

Ionization of Amino Acids

- At low pH, the amino acid exists in a positively charged form (cation).
- At high pH, the amino acid exists in a negatively charged form (anion).
- Between the pK_a for each group, the amino acid exists in a zwitterion form, in which a single molecule has both a positive and negative charge.

Cation → Zwitterion → Anion

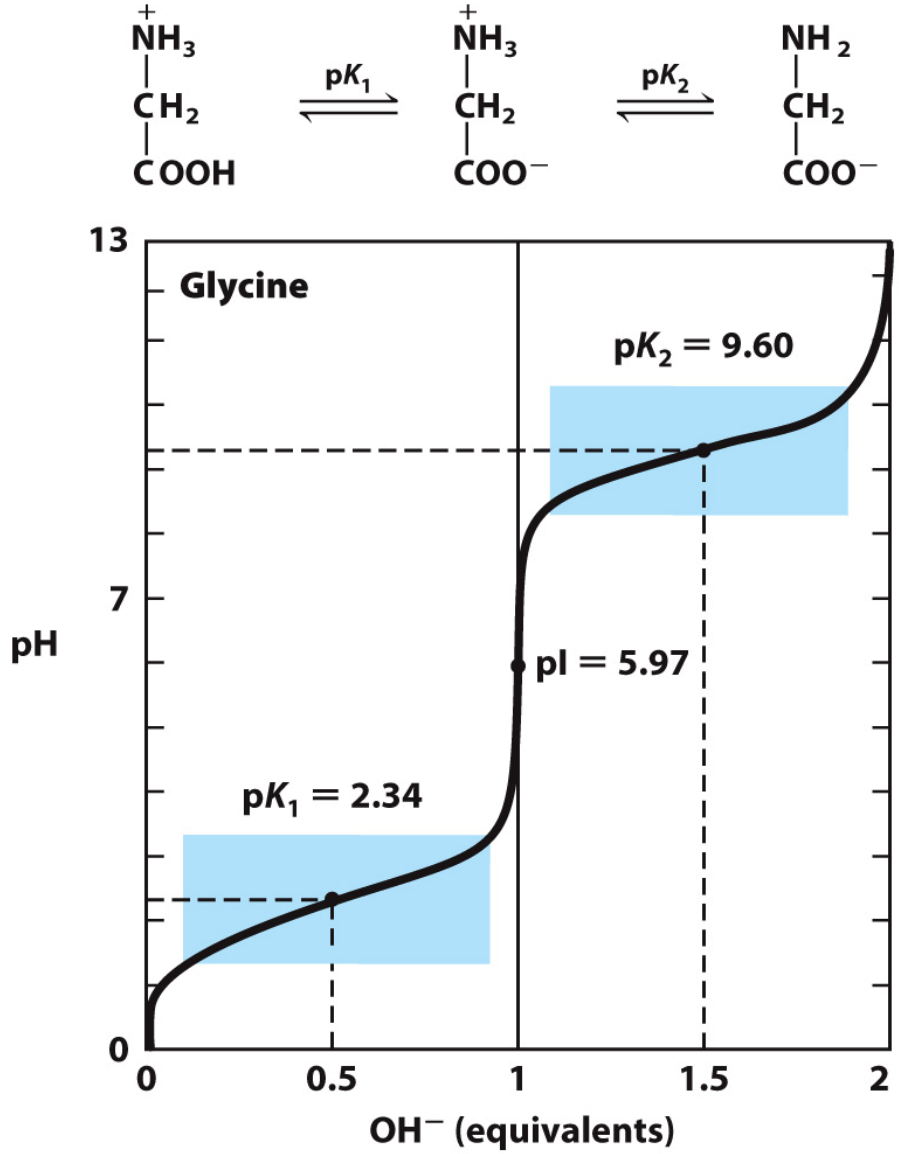


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Amino Acids Carry a Net Charge of Zero at a Specific pH (the pI)

- Zwitterions predominate at pH values between the pK_a values of the amino and carboxyl groups.
- For amino acids without ionizable side chains, the Isoelectric Point (equivalence point, pI) is:

$$pI = \frac{pK_1 + pK_2}{2}$$

- At this point, the net charge is zero.
  - AA is least soluble in water.
  - AA does not migrate in electric field.

Amino Acids Can Act as Buffers

Amino acids with uncharged side chains, such as glycine, have two pK_a values:

- The pK_a of the α-carboxyl group is 2.34.
- The pK_a of the α-amino group is 9.6.

As buffers prevent change in pH close to the pK_a, glycine can act as a buffer in two pH ranges.

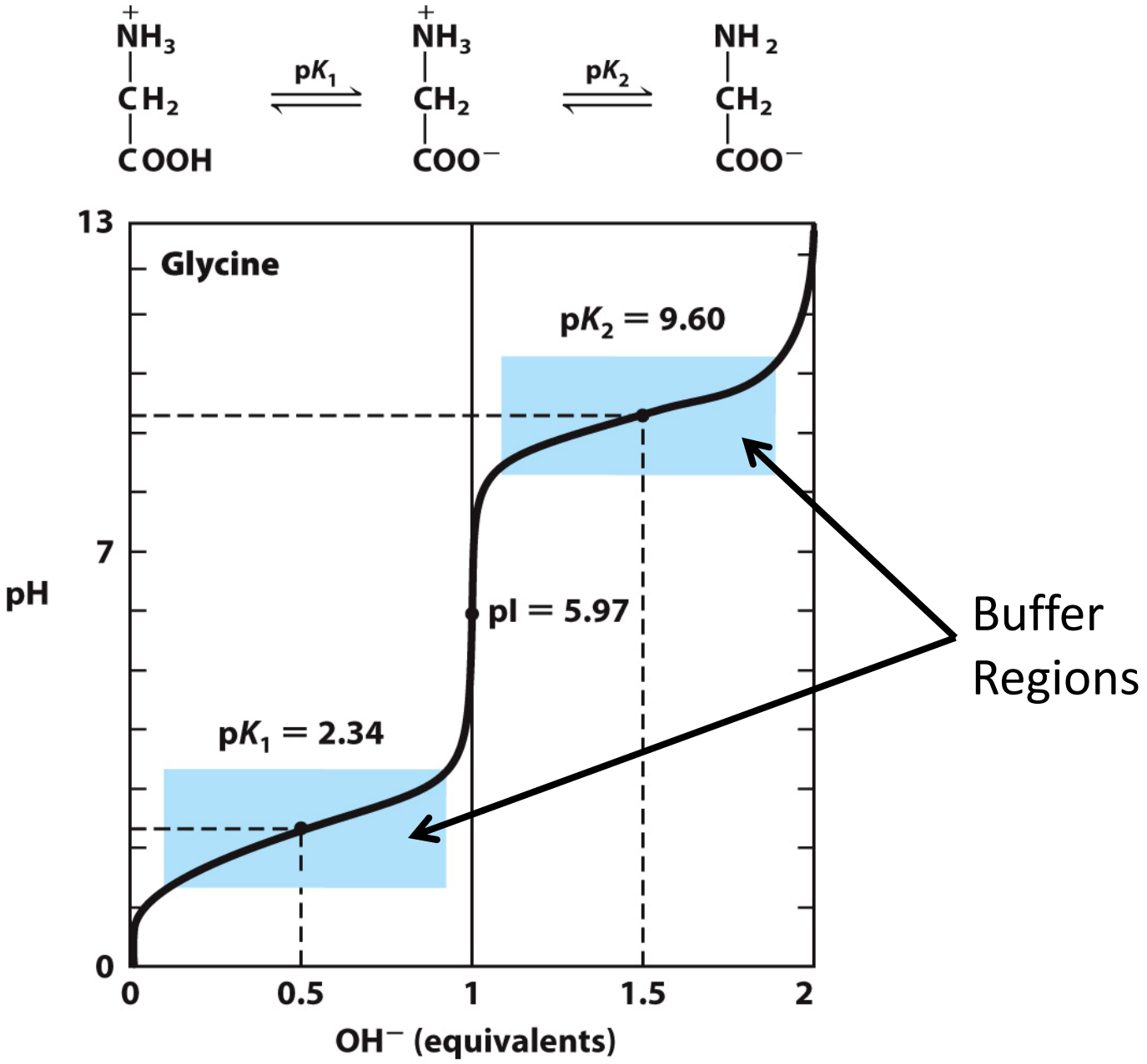


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Ionizable Side Chains Also Have pK_a and Act as Buffers

- Ionizable side chains influence the pI of the amino acid.
- Ionizable side chains can be also titrated.
- Titration curves are now more complex, as each pK_a has a buffering zone of 2 pH units.

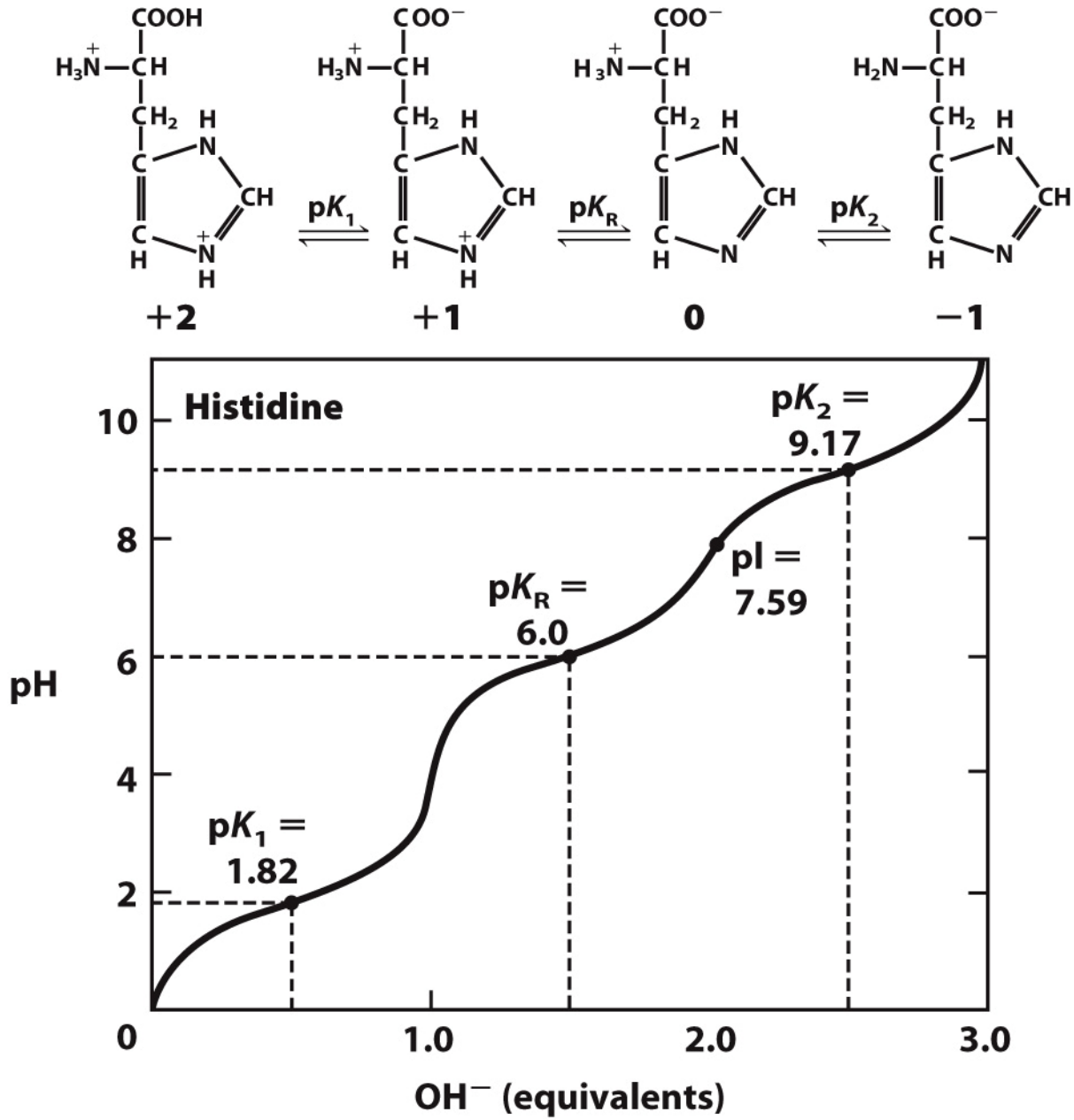
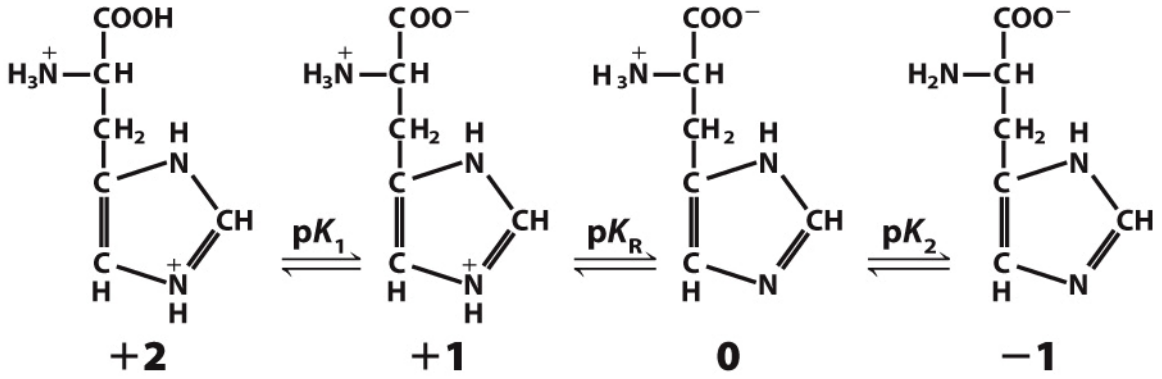


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How to Calculate the pI When the Side Chain Is Ionizable

- Identify species that carries a net zero charge.



- Identify the pK_a value that defines the acid strength of this zwitterion: (pK_R = 6).
- Identify the pK_a value that defines the base strength of this zwitterion: (pK_2=9,17).
- Take the average of these two pK_a values.

What is the pI of histidine? (7.59)

Amino Acids Polymerize to Form Peptides

- Peptides are small condensation products of amino acids.
- They are “small” compared with proteins (M_w < 10 kDa).

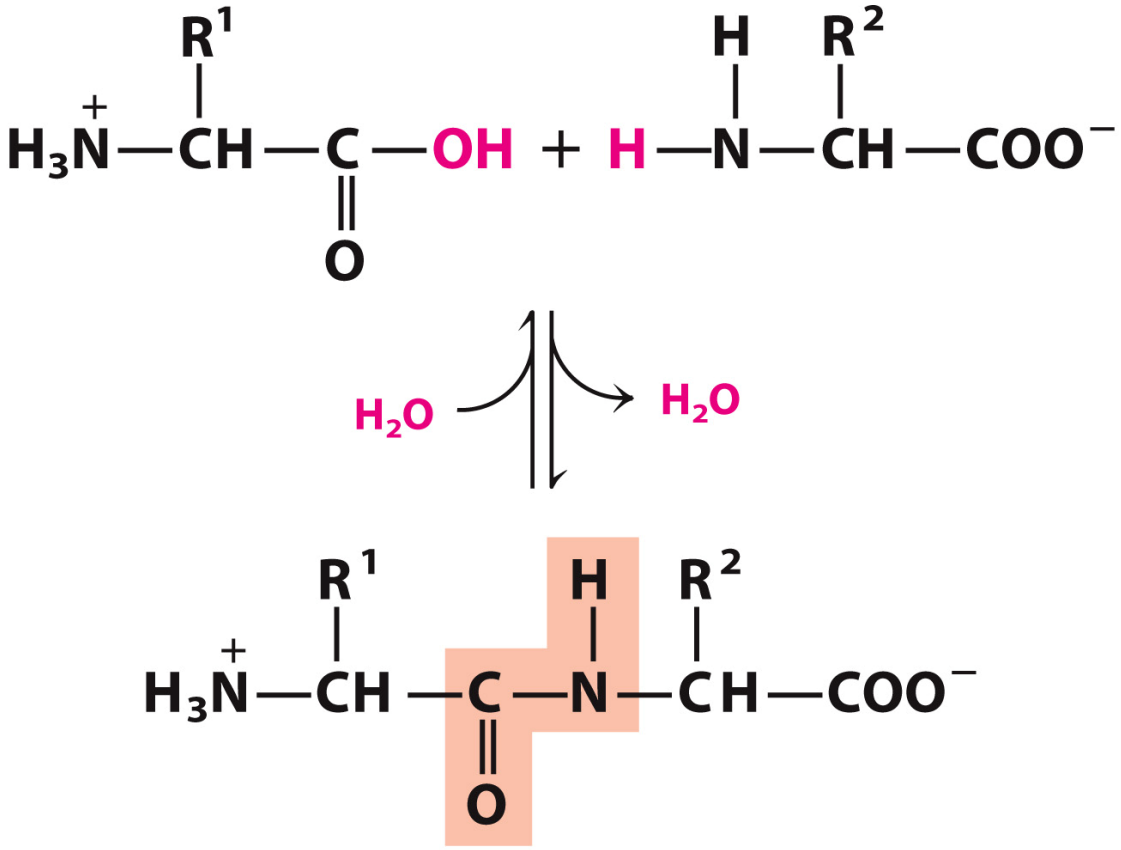


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Peptide Ends Are Not the Same

Numbering (and naming) starts from the amino terminus (N-terminal).

AA_1 AA_2 AA_3 AA_4 AA_5

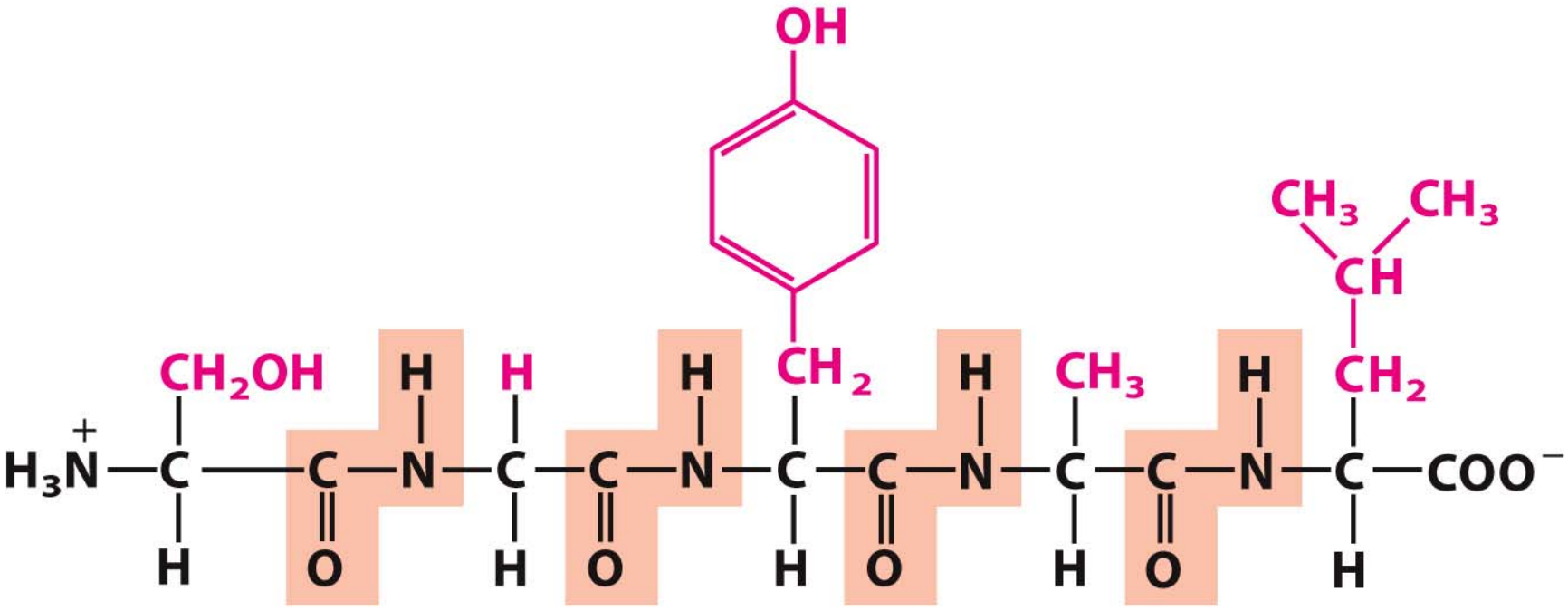


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Naming Peptides: Start at the *N*-terminal

- Using full amino acid names:
  - serylglycyltyrosylalanylleucine
- Using the three-letter code abbreviation:
  - Ser-Gly-Tyr-Ala-Leu
- For longer peptides (like proteins) the one-letter code can be used:
  - SGYAL

Proteins are comprised of:

- Polypeptides (covalently linked α-amino acids) + possibly:
  - cofactors
    - functional non-amino acid component
    - metal ions or organic molecules
  - coenzymes
    - organic cofactors
    - NAD^+ in lactate dehydrogenase
  - prosthetic groups

–covalently attached cofactors

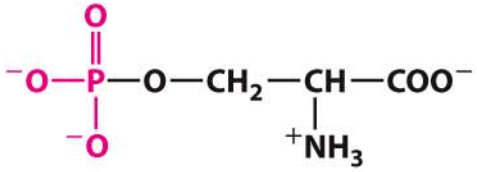
–heme in myoglobin

  - other modifications (posttranslational modifications)

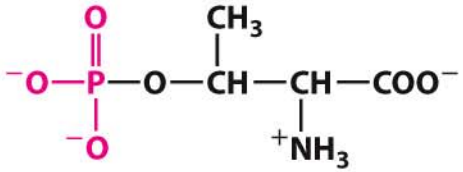
Uncommon Amino Acids in Proteins

- Not incorporated by ribosomes
  - except for selenocysteine
- Arise by posttranslational modifications of proteins
- Reversible modifications, especially phosphorylation, are important in regulation and signaling.

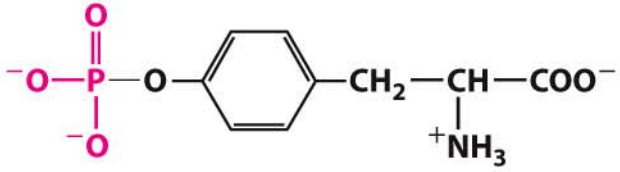
Reversible Modifications of Amino Acids



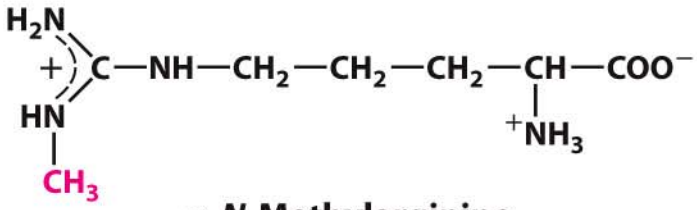
Phosphoserine



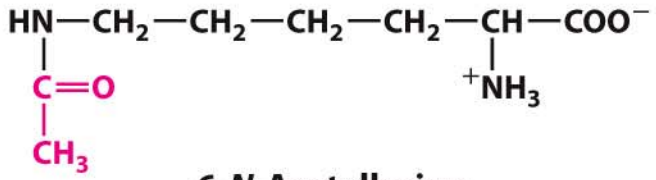
Phosphothreonine



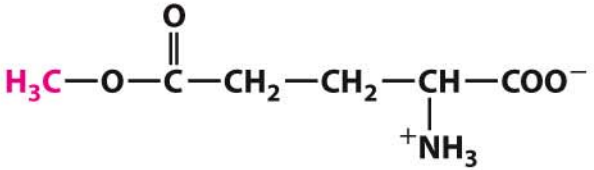
Phosphotyrosine



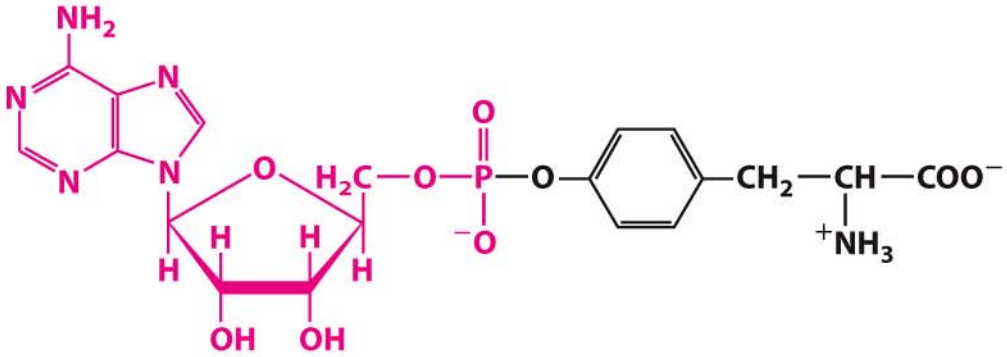
ω-*N*-Methylarginine



6-*N*-Acetyllysine



Glutamate γ-methyl ester



Adenylyltyrosine

Figure 3-8b
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